

WASP-12b

R-0347

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_231206 · created 2026-07-18T09:01:05+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460285.8612 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.113 +/- 0.011
Transit depth (Rp/Rs)^2	1.29 +/- 0.26 [%]
Orbital Inclination (inc)	86.74 +/- 1.06
Airmass coefficient 1 (a1)	0.9831 +/- 0.0071
Airmass coefficient 2 (a2)	0.015 +/- 0.027
Scatter in the residuals of the lightcurve fit is	0.72 %
Best Comparison Star	#2 - [292, 256]
Optimal Aperture	4.42
Optimal Annulus	10.31
Transit Duration (day)	0.1287 +/- 0.0017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.113 ± 0.011 vs 0.1178 (deviation 0.28σ)
Duration fit vs published	0.1287 d vs 0.125 d (deviation 1.39σ)
Mid-time O–C	1573.5 min
Depth SNR	6.6
Residual scatter	0.72 %

Light curve

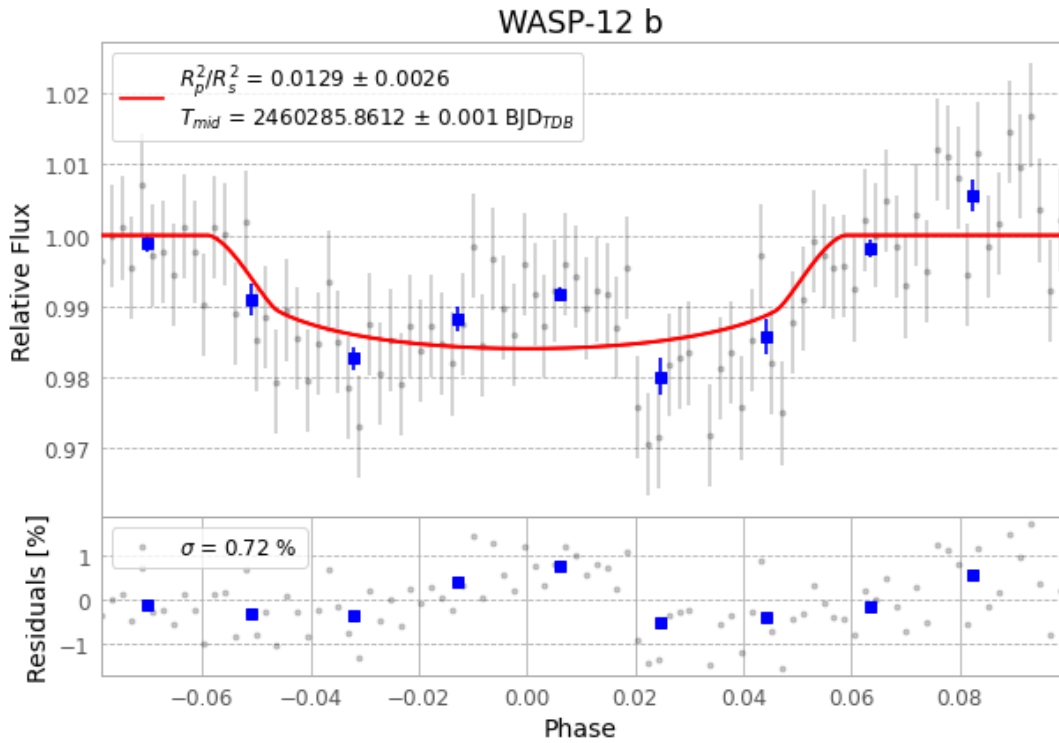


Figure 1 — FinalLightCurve_WASP-12 b_2023-12-06.png (SHA-256 cceb62b9eb270dbd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [310, 225], 2 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 4c008cea7c1c4c24...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8b944da

Data provenance

94 FITS frames (62 MB), WASP-12231207062715.FITS → WASP-12231207110615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-231206.log (b5d7d3618fc9...), FinalLightCurve_WASP-12 b_2023-12-06.png (cceb62b9eb27...), FinalLightCurve_WASP-12 b_2023-12-06.csv (1803d4ac9982...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 09:01Z.